

Class 7 Maths SA1 - Chapters 7 and 8 Practice Question Paper

Scope: Chapter-wise practice papers for triangles and fractions.

SOLVERS MATHEMATICS — CLASS 7

CHAPTER 7: A TALE OF THREE INTERSECTING LINES (TRIANGLES)

CHAPTER ASSESSMENT TEST PAPER

Time Allowed: 1 Hour | **Maximum Marks:** 40 Marks

SECTION A: OBJECTIVE TYPE QUESTIONS (10 Marks)

I. Multiple Choice Questions (1 Mark each)

- The sum of the interior angles of any triangle is always: [1]
 - 90°
 - 180°
 - 270°
 - 360°
- An exterior angle of a triangle is equal to: [1]
 - Sum of its interior opposite angles
 - Sum of all interior angles
 - Difference of its interior opposite angles
 - 180° minus opposite angle
- Which of the following sets of side lengths can form a valid triangle? [1]
 - $2\text{ cm}, 3\text{ cm}, 5\text{ cm}$
 - $3\text{ cm}, 6\text{ cm}, 10\text{ cm}$
 - $6\text{ cm}, 8\text{ cm}, 10\text{ cm}$
 - $4\text{ cm}, 4\text{ cm}, 9\text{ cm}$
- A triangle in which all three sides are of unequal lengths is known as: [1]
 - Equilateral triangle
 - Isosceles triangle
 - Scalene triangle

(d) Right-angled triangle

5. In a right-angled triangle, if one acute angle is 42° , the other acute angle is: [1]

(a) 42°

(b) 48°

(c) 138°

(d) 90°

II. Fill in the Blanks (1 Mark each)

6. The line segment joining a vertex of a triangle to the midpoint of its opposite side is called a . [1]

7. A triangle cannot have more than obtuse angle. [1]

8. In any triangle, the side opposite to the largest angle is always the side. [1]

III. True or False (1 Mark each)

9. The sum of any two sides of a triangle must be strictly greater than the third side. [1]

10. An equilateral triangle has each angle measuring 60° . [1]

SECTION B: VERY SHORT ANSWER QUESTIONS (6 Marks)

11. The angles of a triangle are in the ratio $1 : 2 : 3$. Find the measure of all three angles. [2]

12. In $\triangle ABC$, the exterior angle at vertex C measures 110° . If one interior opposite angle is 45° , find the other interior opposite angle. [2]

13. Can a triangle have angles measuring 65° , 75° , and 50° ? Justify your answer. [2]

SECTION C: SHORT ANSWER QUESTIONS (12 Marks)

14. One of the acute angles of a right-angled triangle is twice the other. Find the measures of both acute angles. [3]

15. In $\triangle PQR$, the bisector of $\angle P$ meets QR at S . If $\angle Q = 70^\circ$ and $\angle R = 30^\circ$, find $\angle PSR$ and $\angle PSQ$. [3]

16. The lengths of two sides of a triangle are 8 cm and 11 cm . Between which two numbers must the length of the third side fall? [3]

17. In the given figure, find the values of the unknown variables x and y :

Exterior angle is 120° , one interior opposite angle is x , the adjacent interior angle is y , and the second interior opposite angle is 50° . [3]

SECTION D: LONG ANSWER QUESTIONS (8 Marks)

18. (a) State and prove the **Angle Sum Property of a Triangle** using a parallel auxiliary line.

(b) In $\triangle ABC$, $\angle B = 2\angle A$ and $\angle C = 3\angle A$. Find all three angles and classify the triangle. [4]

19. Write down the step-by-step procedure for constructing $\triangle ABC$ where side $AB = 6\text{ cm}$, $\angle A = 60^\circ$, and $\angle B = 45^\circ$ using a ruler, protractor, and compass. Calculate the measure of the third angle $\angle C$. [4]

SECTION E: CASE-STUDY / COMPETENCY-BASED QUESTION (4 Marks)

20. Case Study: Solar Panel Roof Framing

A solar installation crew mounts photovoltaic panels on a rooftop triangular truss structure ABC :

- Base beam BC is horizontal.
- Support strut AB makes an inclination of 35° with the horizontal ($\angle B = 35^\circ$).
- Ridge angle at vertex A is designed to be 85° ($\angle A = 85^\circ$).

(a) Calculate the tilt angle of strut AC with the horizontal roofline ($\angle C$). [1]

(b) If an exterior gusset plate at vertex B measures the exterior angle, what is its value? [1]

(c) If strut $AB = 4.2\text{ m}$ and strut $AC = 2.8\text{ m}$, what is the theoretical maximum allowable length for the base beam BC to keep the truss rigid? [2]

SOLVERS MATHEMATICS — CLASS 7

CHAPTER 8: WORKING WITH FRACTIONS

CHAPTER ASSESSMENT TEST PAPER

Time Allowed: 1 Hour | **Maximum Marks:** 40 Marks

SECTION A: OBJECTIVE TYPE QUESTIONS (10 Marks)

I. Multiple Choice Questions (1 Mark each)

1. A fraction whose numerator is greater than its denominator is called a/an: [1]

- (a) Proper fraction
- (b) Improper fraction
- (c) Unit fraction
- (d) Equivalent fraction

2. The reciprocal of $\frac{2}{5}$ is: [1]

- (a) $\frac{5}{13}$
- (b) $\frac{13}{5}$
- (c) $\frac{2}{3}$
- (d) $\frac{5}{6}$

3. The product of $\frac{4}{7} \times \frac{21}{16}$ in simplest form is: [1]

- (a) $\frac{3}{4}$
- (b) $\frac{7}{4}$
- (c) $\frac{4}{3}$
- (d) $\frac{84}{112}$

4. Which of the following fractions is in simplest irreducible form? [1]

- (a) $\frac{9}{21}$
- (b) $\frac{15}{25}$
- (c) $\frac{8}{15}$
- (d) $\frac{14}{49}$

5. What is the value of $\frac{3}{4} \div \frac{9}{8}$? [1]

- (a) $\frac{2}{3}$
- (b) $\frac{3}{2}$
- (c) $\frac{27}{32}$
- (d) $\frac{1}{2}$

II. Fill in the Blanks (1 Mark each)

6. $\frac{3}{8} + \frac{5}{8} = \underline{\hspace{2cm}}$. [1]

7. To divide a fraction by another fraction, we multiply the first fraction by the $\underline{\hspace{2cm}}$ of the second fraction. [1]

8. $\frac{2}{5}$ of 40 kg = $\underline{\hspace{2cm}}$ kg. [1]

III. True or False (1 Mark each)

9. Multiplying a proper fraction by another proper fraction always yields a product smaller than each of the fractions. [1]

10. The fraction $\frac{0}{5}$ has no reciprocal. [1]

SECTION B: VERY SHORT ANSWER QUESTIONS (6 Marks)

11. Convert $\frac{43}{6}$ into a mixed fraction. [2]

12. Find an equivalent fraction of $\frac{3}{7}$ having denominator 35. [2]

13. Compare using $>$, $<$, or $=$: [2]

$$\frac{5}{8} \quad \Box \quad \frac{7}{12}$$

SECTION C: SHORT ANSWER QUESTIONS (12 Marks)

14. Evaluate step-by-step: [3]

$$3\frac{1}{4} + 2\frac{2}{5} - 1\frac{1}{2}$$

15. A rectangular photo frame has length $12\frac{1}{2}$ cm and breadth $8\frac{2}{3}$ cm. Find:

(a) The perimeter of the frame.

(b) The area of the frame. [3]

16. Evaluate: [3]

$$\left(\frac{5}{14} \times \frac{21}{25} \right) \div \frac{9}{10}$$

17. A ribbon of length $15\frac{3}{4}$ m is cut into equal pieces, each of length $1\frac{3}{4}$ m. How many pieces are obtained? [3]

SECTION D: LONG ANSWER QUESTIONS (8 Marks)

18. (a) In a library, $\frac{2}{5}$ of the books are in English, $\frac{1}{3}$ are in Hindi, and the remaining 480 books are in regional languages.

Find the total number of books in the library.

(b) How many books are in English, and how many are in Hindi? [4]

19. (a) Illustrate the multiplication $\frac{2}{3} \times \frac{3}{4}$ using an area partition grid model and calculate the product in simplest form.

(b) Evaluate: [4]

$$5\frac{1}{3} \div \left[2\frac{2}{3} + \left\{ 1\frac{1}{2} \times \left(\frac{4}{5} - \frac{2}{5} \right) \right\} \right]$$

SECTION E: CASE-STUDY / COMPETENCY-BASED QUESTION (4 Marks)

20. Case Study: Community Kitchen Baking Recipe

A community bakery prepares special artisan bread loaves. The standard recipe for one batch requires:

- $3\frac{1}{2}$ cups of whole wheat flour
- $\frac{3}{4}$ cup of warm milk
- $\frac{2}{3}$ cup of natural honey

On Saturday, the master baker decides to make $4\frac{1}{2}$ batches to supply a food festival.

(a) How much warm milk is required for all $4\frac{1}{2}$ batches? [1]

(b) How much honey is required for all $4\frac{1}{2}$ batches? [1]

(c) If the bakery has a 15 sack of wheat flour, is it sufficient to bake all $4\frac{1}{2}$ batches? Calculate the exact surplus or shortage. [2]
